

to large residual errors. PID overcomes this problem encountered in PI controllers. Addition of derivative term improves the loop's transient response. This type of controller is mainly used in applications where large thermal loads must be controlled rapidly and accurately. Fig. 15 shows a typical PID controller interfaced with other building blocks, including temperature sensor, reference generating circuit and error amplifier.

The output stage provides necessary drive power to the TE module. In most cases, electronic systems are designed to operate from a single positive DC voltage supply. Also, in most applications, TE modules need to be driven in bipolar mode to cater for both heating and cooling operations. A commonly used circuit topology to provide bipolar drive to TE modules while operating from a single DC voltage is the half-bridge circuit topology (Fig. 16). The circuit is usually driven at the input by a



Fig. 17: Integrated laser-diode drive and temperature controller

driver amplifier stage with a differential output. Transistors Q1-Q4 and Q2-Q3 conduct alternately to provide bidirectional operation of TE module. A cascade connection of two half-bridge circuits may be used to enhance voltage and current drive capability of the output stage.

TE drive and control modules for laser diode temperature control and stabilisation are commercially available, both as general-purpose benchtop equipment as well as for OEM applications. Even integrated-laser-diode drive and temperature-control modules are available for OEM applica-

tions. Fig. 17 shows a representative photograph.

Present-day military laser range-finders and target designators are largely configured around diode-pumped pulsed solid-state lasers. We have discussed in the preceding paragraphs the building blocks of laser-diode drive and temperature-control electronics. Circuit topologies of laser-diode drivers for different modes of operation of laser diodes, including CW, quasi-CW and pulsed modes and also laser-diode temperature drive and stabilisation, have been discussed. Gas laser electronics is in focus in the concluding part of the article.

To be continued

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